



Faculty of Arts, Humanities, and Social Sciences

School of History and Geography

SAMPLE EXAM AY 2025/26

Module Title:	Landscape Evolution
Module Code:	GY4027
Lecturer/Examiner:	Dr. Breandán Anraoi MacGabhann
External Examiner:	Dr Stephen McCarron
Duration:	1 Hour
% of Total Module Marks:	50%

Instructions to Candidates:

- 1. Answer all questions in Section A**
- 2. Answer one question in Section B**
- 3. You may use ONE printed A4 page of notes**

Section A: Answer all questions

1. Why is a highly mature quartz sandstone unlikely to yield significant quantities of new clay minerals upon further chemical weathering? (3)
2. As a river's flow velocity decreases upon entering a lake, which sediment size fraction will be deposited first, and why? (2)
3. How would the transport of a medium-grained sand grain differ in a fast-flowing river compared to an arid desert dune setting? (5)
4. How can the biosphere affect the evolution of a fluvial landscape? (3)
5. Why are sediments deposited by streams and rivers in upland areas rarely preserved as rocks in the geological record? (4)
6. Describe some of the different ways in which clastic sediments can be transported in glacial environments? (5)
7. Why does aeolian saltation on Mars require significantly higher wind speeds to start grain movement compared to Earth? (6)
8. How might heavy irrigation of a golf course in an arid environment like Las Vegas change chemical weathering rates of the underlying soil? (4)
9. A quartz-cemented sandstone and a calcite-cemented sandstone are subjected to slightly acidic rainfall. Which will maintain its structural integrity longer, and why? (3)
10. The rocks visible on the photograph on the following page are primarily mudstones and siltstones, with some thin volcanic ash layers. Explain why there are no coastal sand dunes present in this landscape? (15)



Section B: Answer two questions

11. Contrast how glacial versus fluvial processes can shape upland landscapes.
How do the fundamentally different processes of ice flow vs. water flow affect the resulting erosional landforms and the characteristics of sediments deposited? **(50)**
12. Explain the concept of accommodation space using examples from specific environments (e.g. fluvial landscapes, coastal landscapes)? **(50)**
13. Discuss how the interplay between tectonic uplift or subsidence, eustatic sea-level change, erosion, and deposition controls the preservation potential of a strandplain beach in the geological record? **(50)**
14. Compare and contrast sediment deposition by ice on Earth and Mars? **(50)**
15. "*Chemical weathering is fundamentally controlled by fluid flow inside rocks at the pore scale.*" Critically evaluate this statement. **(50)**
16. Describe and explain how a river channel morphology can adjust to achieve equilibrium in response to spatial and temporal variations in water flow volume and the sediment load being carried? **(50)**

End of Assessment Paper